Fuseini Aboagye

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Professional Summary

Statistics and Economics graduate with a strong foundation in statistical modelling, probability, statistical inference, quantitative analysis, and empirical research. Skilled in analysing complex datasets, developing data-driven solutions, and translating quantitative findings into actionable insights, with a commitment to evidence-based decision-making. Passionate about applying statistical methods to biomedical and public health research to improve health outcomes and inform policy.

Education

University of Ghana — **Accra, Ghana**

B.A Statistics and Economics*: *GPA 3.91/4.0 — ***January 2022 - September 2025***

- First Class Honors.
- Research: Longitudinal Study of Brain Structure Changes in Alzheimer's Disease Using Machine Learning.
- Top 1% of the graduating class.
- Overall Best Student in Statistics and Economics.

Professional Experience

Ghana Revenue Authority, Ministry of Finance — **Agona Swedru, Ghana**

Assistant Client Service Manager — ***May 2024 - July 2024***

- Resolved 50+ taxpayer inquiries weekly on filings, payments, and refunds, achieving a 95% issue-resolution rate within 3 business days.
- Delivered 10+ public education sessions and one-on-one consultations during the month, improving taxpayer awareness of obligations and available services by 30%, measured via post-session feedback.
- Assisted in addressing and closing 20+ taxpayer complaints and disputes, maintaining a 90% client satisfaction rate through timely and professional service.
- Supported the collection and processing of taxes (income, VAT, corporate), helping to streamline workflows and reduce average processing time per transaction by 15%.

Agona West Municipal Assembly, Ministry of Local Government — **Agona Swedru, Ghana**

Data Analyst — ***March 2024 - May 2024***

- Entered and validated 1,500+ records using R and Excel with a 99% accuracy rate, ensuring reliable datasets for municipal decision-making.
- Collected, organized, and maintained statistical data from 6+ municipal departments, improving accessibility and consistency of records.
- Contributed to the preparation of 5 comprehensive statistical reports and 3 presentations, supporting evidence-based discussions at council meetings and public dissemination.
- Streamlined scheduling and documentation processes for the Municipal Statistician, improving workflow efficiency and reducing administrative delays by 20%.

Allahu Akbar Enterprise — **Gomoa Tarkwa , Ghana**

Sales & Customer Service Associate — ***February 2016 - June 2022***

- Assisted an average of 30+ customers daily with product selection, providing tailored recommendations that improved customer satisfaction and repeat purchases.
- Processed hundreds of transactions weekly, ensuring 100% accuracy in cash handling and sales records over 6 years.
- Resolved 95% of customer inquiries and concerns on first contact, maintaining a reputation for prompt and courteous service.
- Contributed to store organization and stock management, helping to reduce product shortages and improve sales efficiency by 15%.

RESEARCH EXPERIENCE

Thesis

- **Aboagye, F**. (2025, November). *Statistical and Machine Learning Approaches to Exploring Brain Structural Changes in Alzheimer's Disease Progression: A Longitudinal Study* (undergraduate thesis, University of Ghana). PDF
  - Built a longitudinal analysis pipeline for repeated-measures brain-structure data (cleaning, harmonization, and feature engineering).
  - Fit and compared statistical and machine-learning models to characterize progression patterns and support risk/trajectory prediction.
  - Summarized findings into publication-ready figures/tables and a complete thesis report.

Manuscript Submitted/Preprints

- Baidoo, D., **Aboagye, F**, Okeke, O. J., et al (2024, in review). Comparing malaria trends in the Comoros Islands: ARIMA modeling and retrospective analysis. arXiv:2508.16018
  - Cleaned and structured malaria surveillance time-series for retrospective trend and seasonality assessment.
  - Implemented ARIMA model selection and diagnostics; evaluated forecasting accuracy with standard error metrics.
  - Contributed to results interpretation and manuscript writing (methods and results sections).
- Baidoo, D., Amevor, R., **Aboagye, F**., Kubuafor, E., et al (2025, in review). Bayesian Hierarchical Methods for Surveillance of Cervical Dystonia Treatments. arXiv:2508.15762
  - Supported model formulation for Bayesian hierarchical surveillance of treatment outcomes across heterogeneous patient groups.
  - Assisted with posterior analysis and visualization to communicate uncertainty and between-group variability.
  - Helped draft and revise figures/tables and reproducible analysis notes for the manuscript.

Academic Projects

- *The Impact of Technological Advances on Students*
  - Designed data-collection instruments and coordinated team roles for survey administration.
  - Cleaned and analyzed responses to quantify digital literacy and perceived learning impacts.
  - Presented key insights and actionable recommendations for classroom technology integration.
- *The Questionnaires Used in the 2021 Population and Housing Census in Ghana*
  - Reviewed questionnaire modules and mapped variables to measurement objectives and policy-relevant indicators.
  - Assessed potential measurement error, wording effects, and comparability issues across items.
  - Produced a structured write-up summarizing implications for data quality and interpretation.

Presentation

- **Aboagye, F**. (2024, October). Exploring How Multiple Indicator Cluster Surveys Are Designed and Implemented. Presented semester-long research at the University of Ghana Economics Department seminar.
  - Synthesized MICS methodology (sampling, field operations, and quality control) into a clear technical narrative.
  - Developed slides with diagrams/flow and key terminology to explain survey design to a mixed audience.
  - Led Q&A and incorporated feedback into revised materials.
- **Aboagye, F**. (2023, March). The Questionnaires Used in the 2021 Population and Housing Census in Ghana. Presented semester-long research at the University of Ghana Economics class.
  - Analyzed question wording and response options for coverage of key demographic and socioeconomic constructs.
  - Highlighted strengths/limitations for downstream statistical analysis and indicator construction.

- Delivered presentation and produced a concise written summary for classmates.

Talks

- **Aboagye, F**. Intro to R. University of Ghana, Department of Statistics and Actuarial Science. Accra, June 2024.
  - Taught core R workflows (data import, wrangling, visualization) with live coding examples.
  - Created a beginner-friendly handout and practice exercises for independent learning.
- **Aboagye, F**. Intro to Latex. University of Ghana, Department of Statistics and Actuarial Science. Accra, October 2023.
  - Demonstrated LaTeX essentials for academic writing (structure, tables, figures, and math).
  - Showed reference management (BibTeX) and best practices for clean, reproducible manuscripts.

TEACHING EXPERIENCE

Department of Statistics and Actuarial Science, University of Ghana **Accra, Ghana**
Teaching Assistant, Instructor: Dr. Issah Seidu ***October 2025 - current***

- Assisted in the instruction of PHMD 457: Biostatistics and Research Methods for Pharmacy, facilitating tutorials and practical sessions on estimation, hypothesis testing, regression modeling, study design, and statistical inference for health research.
- Assisted in the instruction of Statistical Inference (STAT 333) and Advanced Regression (STAT 445), facilitating tutorials and problem-solving sessions on estimation, hypothesis testing, regression modeling, and statistical decision-making.
- Guided students in the use of statistical methods for data analysis, interpretation of empirical results, and evidence-based conclusions, contributing to improved academic performance.
- Provided individualized academic support to undergraduate students, strengthening their understanding and application of statistical inference and regression techniques in coursework and research projects.
- Collaborated with faculty members in supporting course delivery, grading assignments, and maintaining high academic standards within the Department of Statistics and Actuarial Science.

Home For Hope Preparatory School **Gomoa Tarkwa, Ghana**
Class Instructor ***January 2020 - September 2021***

- Administered and graded weekly tests for 40+ students, ensuring timely feedback and measurable academic progress.
- Strengthened parent–teacher collaboration by holding monthly check-ins, which improved student engagement and accountability.
- Designed and implemented behavior improvement strategies, leading to a significant reduction in classroom disruptions.
- Provided 5+ hours of after-school and weekend tutoring weekly, raising struggling students' performance by an average of one grade level.

ACADEMIC SERVICE, LEADERSHIP, & VOLUNTEERING

2024 - Chairperson, Academic and Scholarship Committee: University of Ghana, Jubilee Hall - International Hostel

2024 - Deputy Chairperson, Academic Committee: University of Ghana Actuarial and Statistics Society

2023 - Chief Clerk, Jubilee Hall Parliament: University of Ghana

2023 - Marshall, Jubilee Hall Parliament: University of Ghana

2021 - Peer Mentor, Compassion International Ghana

2021 - Staff Financial Secretary, Home for Hope Preparatory School: Gomoa West Education District

2020 - Pandemic Educator, Economics Department: Potsin T.I. Ahmadiyya Senior High School

2019|20 - Treasurer, General Arts Student Association: Potsin T.I. Ahmadiyya Senior High School

Awards & Honors

2025 - Best Teaching Assistant Award, Department of Statistics and Actuarial Science

2025 - Sadhu T.L. Vaswani/Indian Association of Ghana Endowment Fund Award for Economics, Overall Best Graduating Student in Economics

2024 - Overall Best Level 200 Student in Economics, Vice-Chancellor's Ceremony in Honour of Academic Excellence for 2022/2023 Academic Year

2024 - Overall Best Level 200 Student in Statistics, School of Physical and Mathematical Sciences Award Ceremony for the 2022/2023 Academic Year

2022 - Fondazione Edu Scholarship - University of Ghana Financial Aid Office

2018|20 - Excellence Club Award, Outstanding academic performance (minimum of 6 A's in 8 courses per semester)

2017 - Inter-School Quiz Champion – Won multiple quiz competitions with the school team at the basic school level

2016 - Inter-Cluster Quiz Champion – Secured top position in regional quiz contests, demonstrating teamwork, critical thinking, and academic excellence.

Professional Development & Conferences

StatFest 2025, American Statistical Association **Virtual**
Mentee ***September 2025 - September 2025***

- Attended 4 graduate preparation workshops focused on admissions, research readiness, and professional development.
- Networked with 30+ statisticians and data scientists from academia, industry, and government, expanding professional connections.
- Participated in 1:1 mentoring sessions, receiving tailored guidance on graduate school success and career planning.
- Explored emerging opportunities in statistics and data science, aligning insights with future academic and professional goals.

Highlighted Skills

Soft Skills: Strong communication skills, leadership and team coordination, analytical thinking and problem-solving, collaboration and interpersonal effectiveness, relationship development and networking

Technical Skills: Proficiency in data analysis tools (R, Python), LaTeX, Microsoft 365, Stata, Prompt expert with LLMs

Professional Member Association

American Statistical Association

Institute of Mathematical Statistics

International Society for Data Science and Analytics